

Stat 517
Homework # 4

1. Let $Y \sim \text{Uniform}[0, 1]$ and $X_n = Y^n$. Prove X_n converges in probability to zero.
2. Let the random variable Y_n have a distribution that is $b(n, p)$.
 - (a) Prove that Y_n/n converges in probability to p .
 - (b) Prove that $1 - Y_n/n$ converges in probability to $1 - p$.
 - (c) Prove that $(Y_n/n)(1 - Y_n/n)$ converges in probability to $p(1 - p)$.
3. Let W_n denote a random variable with mean μ and variance b/n^p , where $p > 0$, μ and b are constants (not function of n). Prove that W_n converges in probability to μ .
4. Let $X_1, \dots, X_n$ be iid random variables with common pdf

$$f(x) = \begin{cases} e^{-(x-\theta)} & x > \theta, -\infty < \theta < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

This pdf is called the shifted exponential. Let $Y_n = \min \{X_1, \dots, X_n\}$. Prove that $Y_n \rightarrow \theta$ in probability by first obtaining the cdf of Y_n .

5. Let Y_1 denote the minimum of a random sample size n from a distribution that has pdf $f(x) = e^{-(x-\theta)}$, $\theta < x < \infty$, zero elsewhere. Let $Z_n = n(Y_1 - \theta)$. Investigate the limiting distribution of Z_n .
6. Let Y_n denote the maximum of a random sample of size n from a distribution of the continuous type that has cdf $F(x)$ and pdf $f(x) = F'(x)$. Find the limiting distribution of $Z_n = n[1 - F(Y_n)]$.
7. Let Y be $b(72, 1/3)$. Approximate $P(22 \leq Y \leq 28)$.
8. Let $f(x) = 1/x^2$, $1 < x < \infty$, zero elsewhere, be the pdf of a random variable X . Consider a random sample of size 72 from the distribution having this pdf. Compute approximately the probability that more than 50 of the observations of the random sample are less than 3.